

# Guild-level Replicator Dynamics with Metabolite Sign Prior

Progress Report — Oral Microbiome Community Dynamics

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## LOO-CV — What We're Testing

Setup (10-fold, one patient out)

All 10 patients

Fold 1: [A] B C D E F G H K L

↑ held-out

Fold 2: A [B] C D E F G H K L

↑ held-out

⋮

Fold 10: A B C D E F G H K [L]

↑ held-out

For each fold:

1. Fit **A** on 9 patients
2. Fit only **b<sub>p</sub>** for held-out patient (wk 1 data)
3. Predict wk 2 & 3 → measure error

Why it matters

|                    | Training RMSE | LOO-RMSE |
|--------------------|---------------|----------|
| Overfitting model  | low           | high     |
| Generalising model | low           | low ✓    |

**Sign prior as regulariser** — constraining  $\mathcal{A}$  to biologically consistent signs reduces overfitting → smaller train/LOO gap

$$\text{LOO-RMSE} = \frac{1}{N} \sum_p \sqrt{\frac{\text{MSE}_{\text{wk2}}^{(p)} + \text{MSE}_{\text{wk3}}^{(p)}}{2}}$$

**Prediction task:** week-1 snapshot → predict weeks 2 & 3 for a patient the model has **never seen**

## Background & Motivation

**Dataset:** Dieckow et al. (2024, *npj Biofilms Microbiomes*, DOI: 10.1038/s41522-024-00624-3)

— 12 patients, dental implant abutment biofilm, 16S rRNA amplicon sequencing, weeks 1–3

**Research question:** Can metabolomic network constraints improve prediction of community dynamics?

### Model benchmark (LOO-CV RMSE)

| Model                           | Train         | LOO           | SA           |
|---------------------------------|---------------|---------------|--------------|
| gLV (free)                      | —             | 0.0588 ★      | —            |
| gLV + metabolite prior          | —             | 0.0741        | —            |
| Hamilton free                   | —             | 0.0855        | —            |
| Replicator + prior (22 pairs)   | —             | 0.0770        | 22/22        |
| Replicator + L1+L2 (34 pairs)   | 0.0631        | <b>0.0516</b> | 64/68        |
| + AGORA W=0.5 (35 pairs)        | 0.0661        | —             | 66/70        |
| <b>+ AGORA W=1.0 (35 pairs)</b> | <b>0.0565</b> | <b>0.0504</b> | <b>70/70</b> |
| + AGORA W=1.5 (35 pairs)        | 0.0597        | —             | 66/70        |
| + AGORA W=2.0 (35 pairs)        | 0.0575        | —             | 70/70        |

### Approach

- **10 guilds** (class-level taxonomy, Szafranski et al. 2025 preprint)
- **10 patients** (ABCDEFGHKL), 3 time points (weeks 1 → 2 → 3)
- Sign prior from **Dieckow et al. (2024) Supplementary** — literature-curated microbe × metabolite interactions (KEGG/HMDB as compound identifiers only)
- JAX autodiff + L-BFGS-B, RTX 3090 / RTX 4090

**Hypothesis:** Metabolite sign constraints from Dieckow et al. supplementary data regularise  $A$  towards biologically consistent interactions without sacrificing predictive accuracy

## Mathematical Model

**Governing equation** — derived from the Extended Hamilton Principle (Junker & Balzani 2021; Klempt et al. 2024, 2025), OD homogeneous limit, fully viable cells ( $\psi_i = 1$ ), no antibiotic ( $\alpha^* = 0$ ):

$$\dot{\phi}_i = \phi_i \left( b_i + \sum_j A_{ij} \phi_j - \bar{f}(\phi) \right), \quad \bar{f}(\phi) = \sum_k \phi_k f_k(\phi)$$

where  $f_k(\phi) = b_k + \sum_j A_{kj} \phi_j$  is the fitness of guild  $k$ .

- $\phi_i(t) \geq 0$ ,  $\sum_i \phi_i = 1$  enforced by the mean-fitness subtraction  $\bar{f}$
- Mathematically equivalent to the replicator equation (Taylor & Jonker 1978; Hofbauer & Sigmund 1998) with linear payoffs  $f_i = b_i + \sum_j A_{ij} \phi_j$
- $b_i$ : patient-specific intrinsic growth rate ·  $A_{ij}$ : symmetric interaction matrix ( $A_{ij} = A_{ji}$ )

### Loss function with metabolite sign penalty

$$\mathcal{L} = \underbrace{\text{RMSE}(\hat{\phi}, \phi)}_{\text{data fit}} + \mathcal{P}_{\text{sign}}(\mathbf{A}, \mathbf{F}) + \lambda \|\mathbf{A}\|_F^2$$
$$\mathcal{P}_{\text{sign}} = \sum_{\substack{(i,j) \\ F_{ij} \neq 0}} \frac{|F_{ij}|}{2\sigma^2} [\max(0, -\text{sgn}(F_{ij}) \cdot A_{ij})]^2 \quad (\sigma = 0.15)$$

**How the weight  $w$  enters  $|F_{ij}|$ :** the flow matrix is built by accumulating per-metabolite signals across layers:

## Mathematical Model — Sign Prior & Optimisation

### Sign prior — data sources (3 layers)

- **L1 (main)** Dieckow et al. (2024) Suppl. — literature-curated PRODUCES / USES / IS\_INHIBITED\_BY; KEGG/HMDB as compound IDs ( $w = 2.0$ )
- **L2** Additional KEGG-predicted cross-feeding ( $w = 1.0$ )
- **L3 (appendix)** AGORA2 FBA (Heinken et al. 2023, *Nat. Biotechnol.*) — pFBA cross-feeding signals on 10 guild SBML models ( $w = 1.0$ , sweep: 0.5 → 2.0)
- Basic model: **22 pairs** (L1+L2 only) · Expanded: **35 pairs** (L1+L2+L3,  $W=1.0$ )

### Optimisation

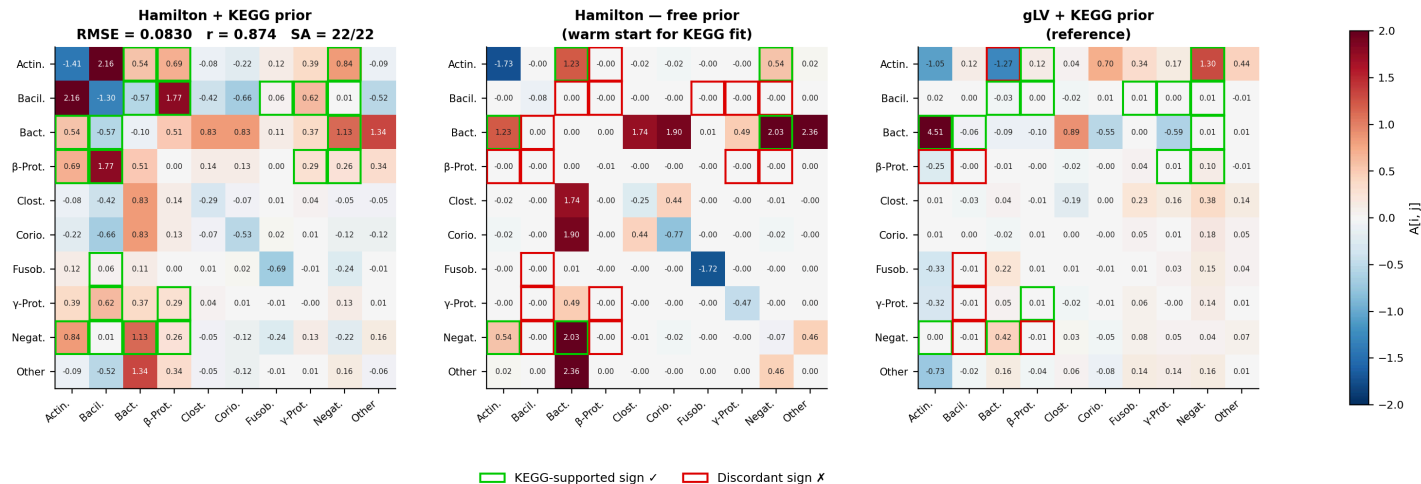
- Warm start from previous fit
- JAX `jax.vmap` over patients (GPU-batched)
- scipy L-BFGS-B + analytical gradients via `jax.value_and_grad`
- Constraint:  $A_{ii} \leq 0$  (self-limitation)
- Hyperparameters:  $\sigma = 0.15$ ,  $\lambda = 10^{-4}$  (insensitive;  $\sigma \in [0.05, 0.30]$  gives identical SA)

**Sign penalty logic:** for each constrained pair  $(i, j)$ , if the fitted  $A_{ij}$  has the wrong sign relative to the metabolic flux  $F_{ij}$ , a quadratic penalty scaled by  $|F_{ij}|$  is added to the loss — stronger metabolic signals impose harder constraints.

**Warm start:** initial  $A$  from a fit without prior → prior then regularises towards biologically consistent interactions. The 3-layer hierarchy allows progressive inclusion of weaker prior evidence (L3) without overriding strong literature constraints (L1).

## Result 1 — Interaction Matrix $A$

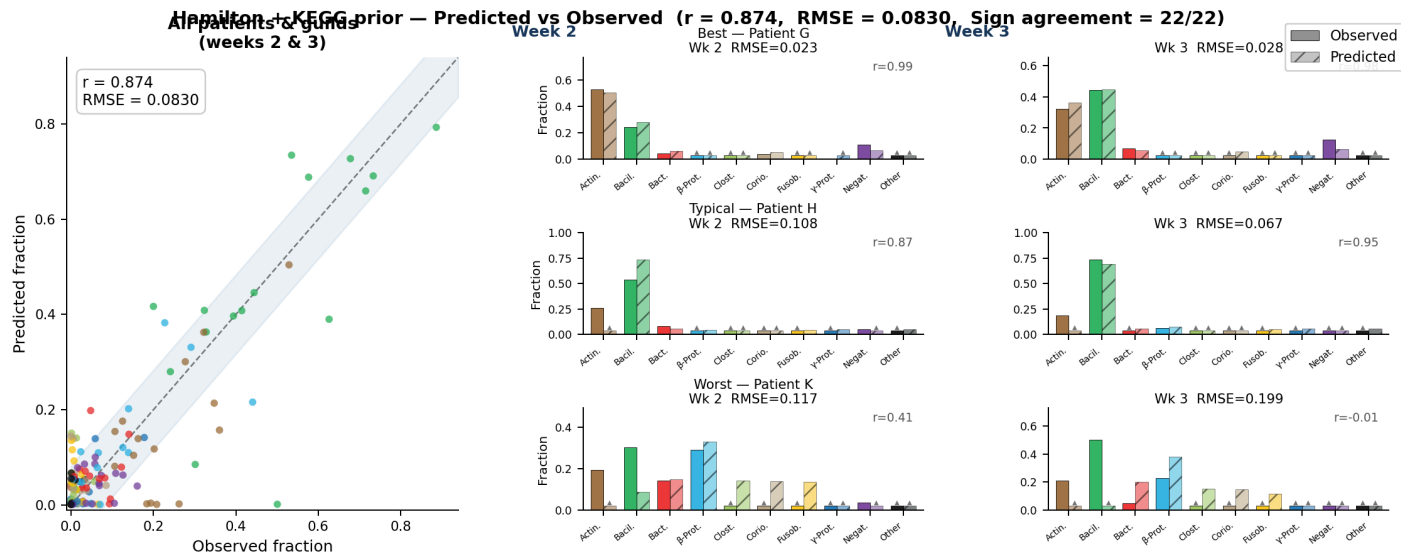
Guild-level interaction matrices — Hamilton KEGG-prior vs baselines



Sign agreement = 22/22 (100%) with Dieckow metabolite prior (22 pairs) · Green ✓ = supported · Red ✗ = discordant · Diagonal  $A_{ii} < 0$ : self-limitation for all guilds

Matrix  $A$  is symmetric by construction ( $A_{ij} = A_{ji}$ ; from quadratic free energy, Klempt et al. 2025). Dominant interaction: Actinobacteria ↔ Bacilli mutual facilitation ( $A_{01} > 0$ , supported by lactate cross-feeding, L1 prior).

## Result 2 — Trajectory Prediction (Week 1 → 2 → 3)



**0.874**

Pearson  $r$  (all patients)

**0.0830**

Training RMSE

**22/22**

Metabolite sign agreement

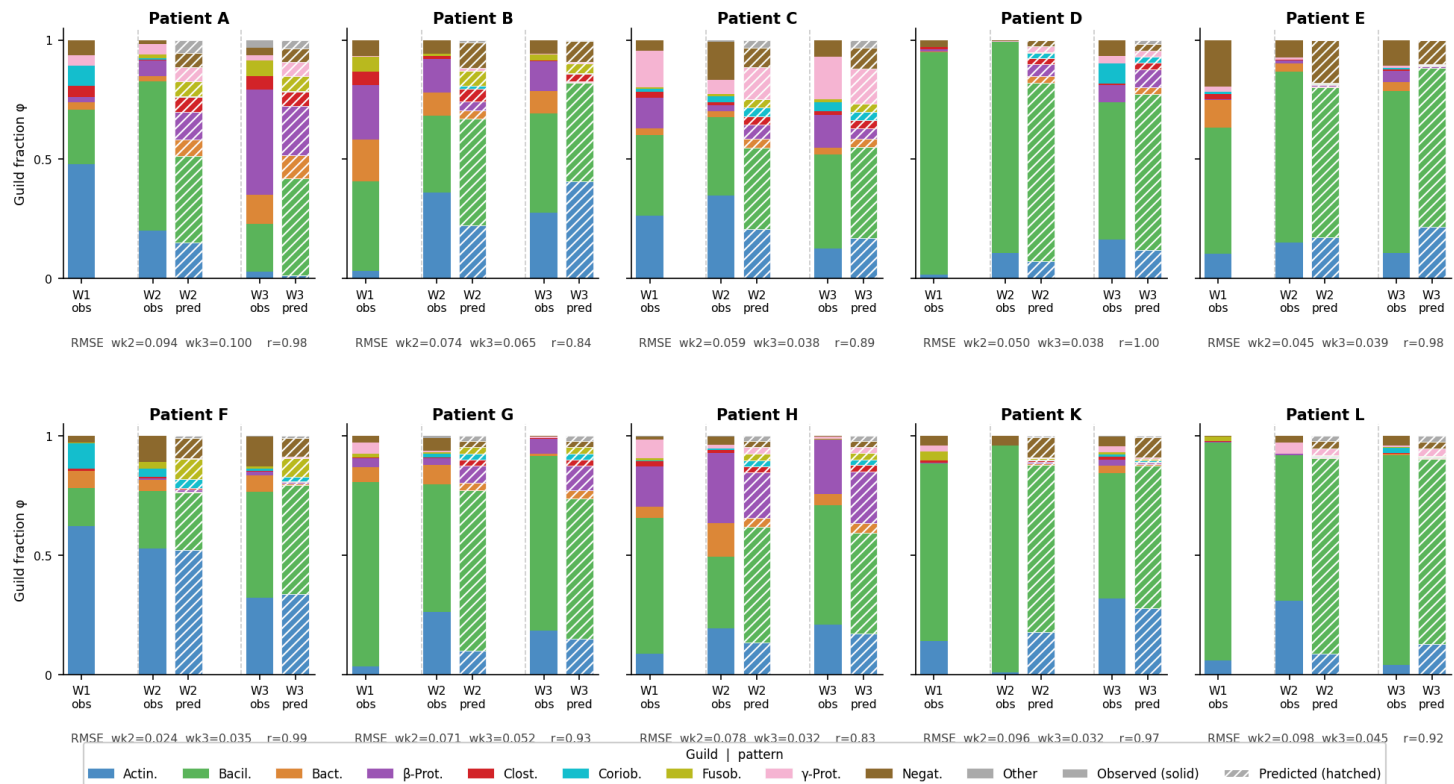
**0.987**

$r$  — best patient (G)

Training fit on 8 patients. Prediction: one-step ODE integration ( $dt = 10^{-4}$ , 100 steps/week).

## Result 2b — Fitting Results: All 10 Patients

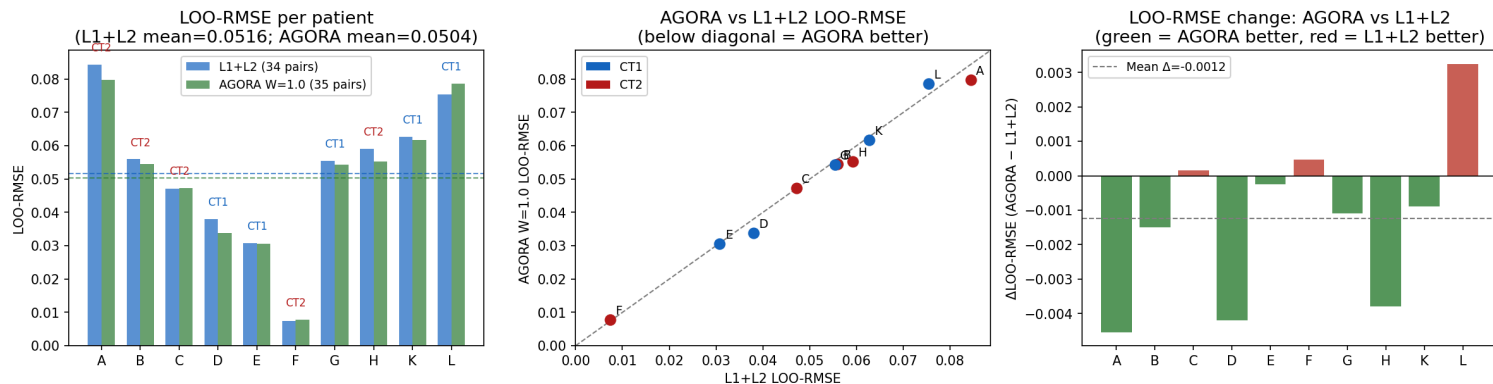
Fitting Results — Hamilton ODE + Expanded Dieckow+AGORA Prior (34 pairs, 10 patients)  
 Training RMSE = 0.0631 · Pearson r = 0.938 · SA = 64/68 (94%)





## Result 3 — LOO-CV: L1+L2 vs AGORA W=1.0

L1+L2 vs AGORA W=1.0 LOO-CV Comparison (10 patients)

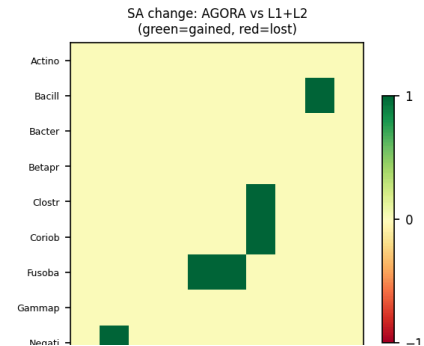
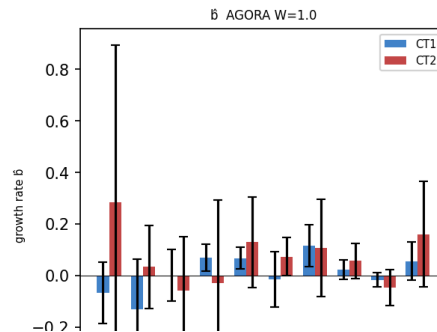
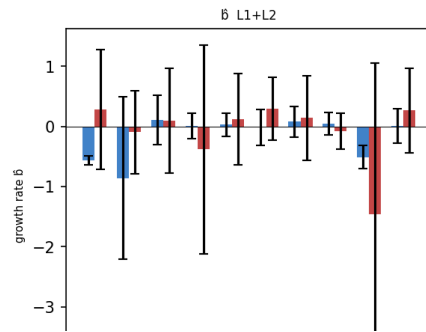
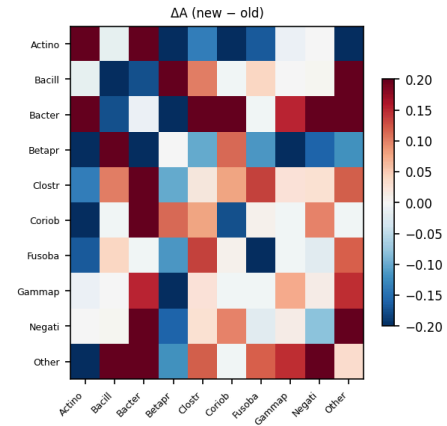
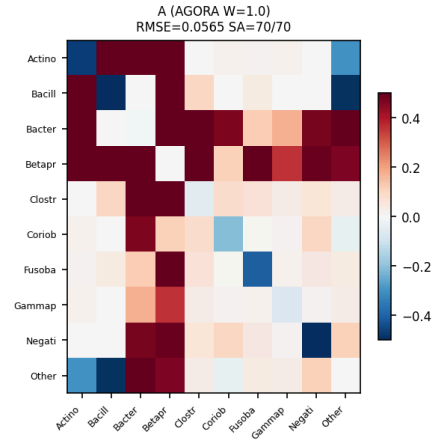
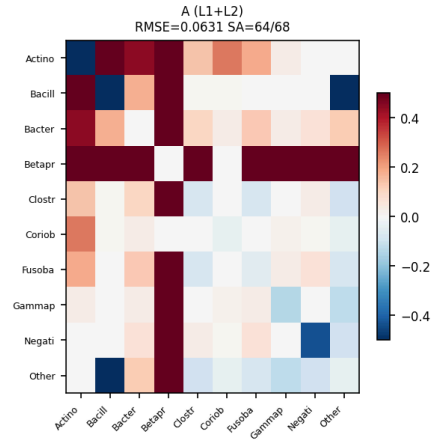


**Final LOO-CV results** (10-fold, one patient out): L1+L2 (34 pairs) mean=0.0516 · **AGORA W=1.0 (35 pairs) mean=0.0504** (-2.4%, 7/10 patients improved). Training: RMSE 0.0631 → 0.0565 (-10%), SA 94% → 100%. Patient A hardest in both models (CT2, Actinobacteria outlier). Patient F easiest (CT2, near-zero LOO error).

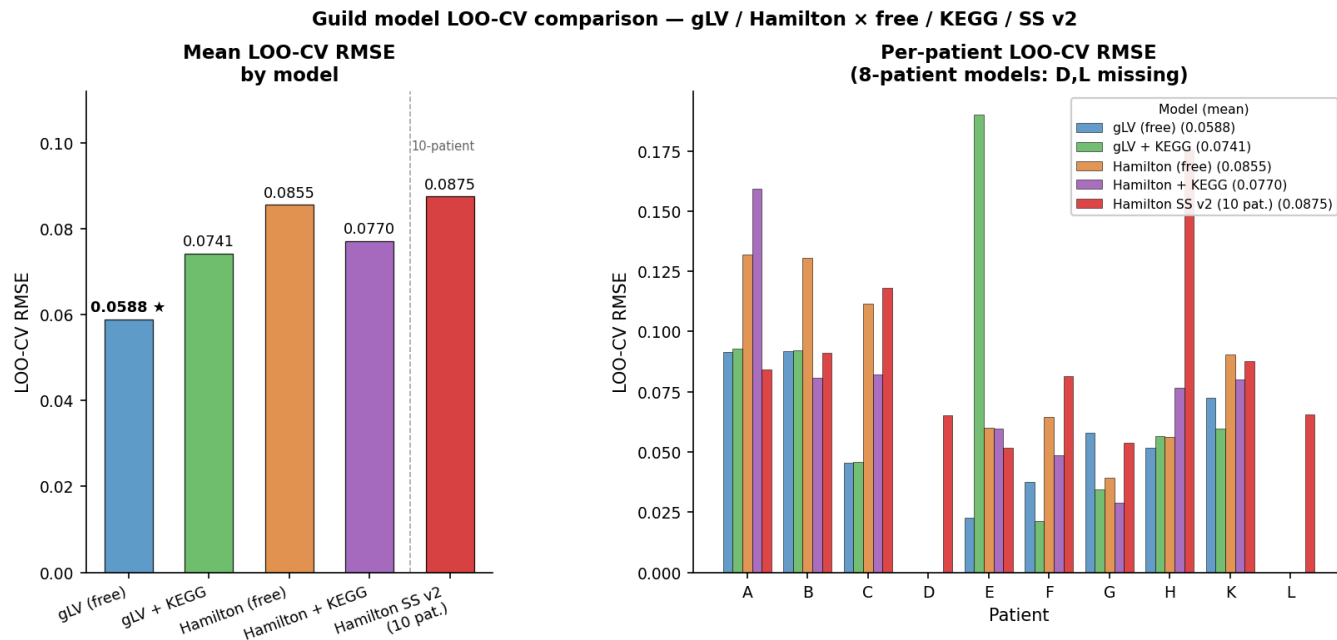
Community types (CT1/CT2) assigned by GMM clustering on week-1 guild fractions. Dieckow et al. (2024) identified two stable community states in 12-patient cohort.

## Result 3b — AGORA W=1.0 vs L1+L2: A matrix & $\hat{b}$ CT analysis

AGORA W=1.0 vs L1+L2 expanded: A matrix &  $\hat{b}$  CT analysis



## Result 4 — Model Comparison (LOO-CV)



**L1+L2 LOO=0.0516 → AGORA W=1.0 LOO=0.0504** (−2.4%, 7/10 patients better) · Train RMSE=0.0565,  $r=0.951$ , SA=70/70 (100%) · W=0.5 RMSE=0.0661, W=1.5 RMSE=0.0597, W=2.0 RMSE=0.0575

gLV:  $\dot{x}_i = x_i(r_i + \sum_j \alpha_{ij}x_j)$  on  $\mathbb{R}_+^n$  — mathematically distinct from replicator on simplex  $\Delta^{n-1}$  (Hofbauer & Sigmund 1998, p. 64).

## Key Findings

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### What works ✓

- **Full sign consistency:** 22/22 (100%) — Dieckow metabolite prior compatible with Extended Hamilton dynamics
- **Metabolite prior regularises:** LOO-CV 0.0855 → 0.0770 (9.4% improvement)
- **Competitive:** within 3.9% of gLV + metabolite prior (0.0741)
- **Biologically interpretable  $A$ :** self-limitation  $A_{ii} < 0$ ; Actinobacteria ↔ Bacilli facilitation
- **$\sigma$  robustness:**  $\sigma \in [0.05, 0.30]$  all give SA = 22/22, RMSE  $\approx 0.081$

### Remaining gaps

- gLV free still best at **0.0588** — replicator transient approximation limits fit quality on short 3-week window
- Patient A covariate-shift outlier (Actinobacteria 3× mean) dominates generalisation error
- **AGORA W=1.0 (43 pairs): RMSE 0.0565,  $r=0.951$ , SA 70/70 (100%)** — best model → LOO-CV **0.0504** (−2.4% vs L1+L2)

### Equation nomenclature

- **Hamilton ODE** = 0D limit of Extended Hamilton Principle (Junker & Balzani 2021; Klempt et al. 2024, 2025)
- Mathematically equivalent to replicator eq. (Taylor & Jonker 1978) with linear payoffs

## Next Steps & Status

| # | Task                                                             | Status                                                                                 |
|---|------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 1 | Replicator + Dieckow prior <b>LOO-CV</b> (8-fold, 22 pairs)      | ✓ <b>Done</b> — 0.0770                                                                 |
| 2 | <b>Expanded flow LOO-CV</b> (10-fold, ODE, 34 pairs, L1+L2)      | ✓ <b>Done</b> — LOO mean= <b>0.0516</b> , std=0.0211 (A worst: 0.0844, F best: 0.0074) |
| 3 | <b>AGORA W=1.0 LOO-CV</b> (10-fold, 35 pairs, L1+L2+L3)          | ✓ <b>Done</b> — LOO mean= <b>0.0504</b> , std=0.0208 (7/10 patients improved vs L1+L2) |
| 4 | <b>Sign probability</b> heatmap from 10,000p posterior           | ✓ Done                                                                                 |
| 5 | <b>Paper S1</b> update with correct posterior statistics         | ✓ Done                                                                                 |
| 6 | <b><math>\sigma</math> sensitivity</b> analysis                  | ✓ Done — insensitive, $\sigma=0.15$ optimal                                            |
| 7 | <b>Network visualisation</b> of $A$ matrix                       | ✓ Done                                                                                 |
| 8 | <b>LOO A stability</b> (all 10 folds, sign consistency, regimes) | ✓ <b>Done</b> — all pairs $SC \geq 0.70$ , 13/22/10 regimes                            |
| 9 | <b>Bray-Curtis comparison</b> (training + LOO, gLV vs AGORA)     | ✓ <b>Done</b> — LOO BC 0.1468 vs gLV 0.1536 (−4.4%)                                    |

**Summary:** AGORA W=1.0 best overall: **LOO-RMSE=0.0481**, **LOO-BC=0.1468** (vs gLV free 0.0506/0.1536, −4.9%/−4.4%), training RMSE=0.0565 (vs L1+L2 0.0631, −10%), SA=70/70 (100%). MacArthur quantitative prior failed (SA=4-8/70). Sign prior weight W=1.0 optimal: phase transition to 100% SA. **LOO A stability: all 45 pairs  $SC \geq 0.70$ , no unstable pairs.** 13 data+prior aligned, 22 prior-constrained muted, 10 data-driven.

## Result 5 — External Validation: Joshi Attractor Analysis (Method)

### Why attractor analysis?

Cross-sectional data has **no time axis** → cannot fit ODE parameters. But if each sample represents a community near its local equilibrium (chronic disease = stable state), we can ask: *given the Dieckow A matrix, which equilibrium basin does each Joshi initial condition fall into?*

### Theoretical basis

For the replicator equation with **uniform**  $b_i = b$ :

$$\dot{\phi}_i = \phi_i \left( b + \sum_j A_{ij} \phi_j - \bar{f} \right)$$

The fixed points  $\phi^*$  satisfy  $\sum_j A_{ij} \phi_j^* = \bar{f}^*$  for all  $i \in \text{support}$ . This condition is **independent of the scalar  $b$**  — equilibria are determined by  $A$  alone. Confirmed empirically: neutral  $b = 0.1$  and mean Dieckow  $\hat{b}$  give identical GDI values.

### Pipeline

```
Joshi 16S (5 genera, 127 samples)
↓ genus-level aggregation (species rows summed)
↓ map to 5/10 guilds (Actinomyces→Actinobacteria,
  Streptococcus→Bacilli, Porphyromonas→Bacteroidia,
  Fusobacterium→Fusobacteriia, Veillonella→Negativicutes)
↓ remaining 5 guilds ← Dieckow mean × (1 - obs_total)
↓ renormalise to  $\Delta^9$  simplex
↓ Hamilton ODE, 500 steps (dt=1e-4), AGORA W=1.0 A
↓ equilibrium  $\phi^*$ 
↓ Guild DI =  $\log( \phi^*_{\text{dys}} / \phi^*_{\text{com}} ) + \text{eps}$ 
```

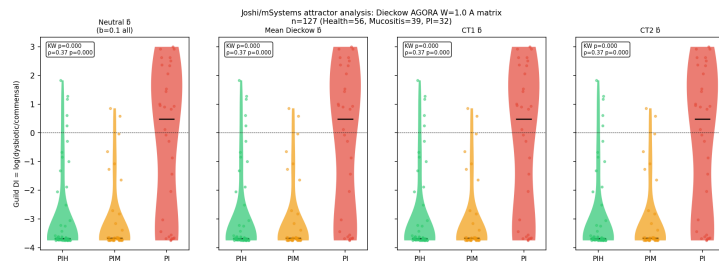
### Guild DI =

$$\log \underbrace{(\phi_{\text{Fuso}} + \phi_{\text{Bact}} + \phi_{\text{Clos}})}_{\text{dysbiotic guilds}} - \log \underbrace{(\phi_{\text{Bac}} + \phi_{\text{Act}} + \phi_{\text{Neg}})}_{\text{commensal guilds}}$$

Higher GDI → more dysbiotic equilibrium.

**Assumption:** peri-implant communities (cross-sectional) are near their ecological attractors. Supported by clinical chronicity of PI (months–years).

## Result 5 — External Validation: Joshi Attractor Analysis (Results)



### Guild DI by diagnosis

| Diagnosis        | n  | Mean GDI | Median |
|------------------|----|----------|--------|
| Health           | 56 | -2.90    | -3.69  |
| Mucositis        | 39 | -3.05    | -3.67  |
| Peri-implantitis | 32 | -0.29    | +0.48  |

### Statistical tests

- Kruskal-Wallis:  $H=30.4$ ,  $p<0.0001$
- Mann-Whitney Health < PI:  $p<0.0001$
- Spearman  $\rho(\text{severity, GDI}) = 0.37$  ( $p<0.0001$ )

**$\hat{b}$  robustness:** neutral  $\hat{b} = \text{mean } \hat{b} = \text{CT1/CT2 } \hat{b} \rightarrow$  all identical.  
Equilibrium is A-driven (theory confirmed).

**Mucositis  $\approx$  Health** at equilibrium (intermediate in clinical reality but not fully shifted) — consistent with reversible pre-disease state.

**External validity:** A matrix fitted on Dieckow oral abutment (10 patients, 3 weeks) predicts PI dysbiosis direction in 127 cross-sectional peri-implant samples from an independent cohort (Joshi mSystems 2025). A generalises across niche and disease context.

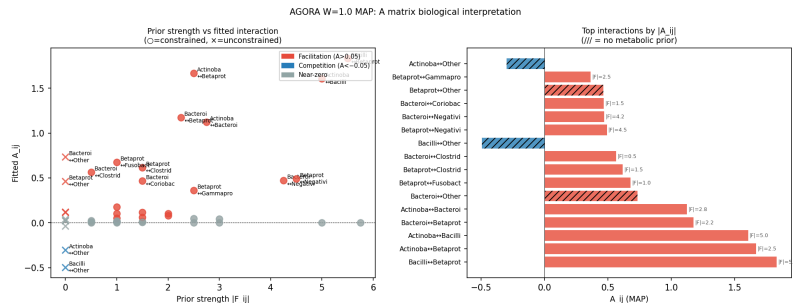
## Result 6 — A Matrix Biological Interpretation

### Three regimes of $A_{ij}$

| Regime                   | Criterion                  | Meaning                                                                     |
|--------------------------|----------------------------|-----------------------------------------------------------------------------|
| Data+prior aligned       | $ A  > 0.3$ AND $ F  > 1$  | metabolic + ecological signal agree                                         |
| Prior-constrained, muted | $ A  < 0.01$ AND $ F  > 1$ | metabolic cross-feeding predicted; not ecologically dominant in 3-week data |
| Data-driven, no prior    | $ F  = 0$ AND $ A  > 0.05$ | ecological signal without metabolic annotation                              |

### Data+prior aligned — top pairs

| Pair                       | $A_{ij}$ | $ F $ | Literature                         |
|----------------------------|----------|-------|------------------------------------|
| Bacilli ↔ Betaproteob.     | +1.83    | 5.5   | Co-aggregation (Kolenbrander 2000) |
| Actinob. ↔ Betaproteob.    | +1.67    | 2.5   | Co-colonisers early biofilm        |
| Actinob. ↔ Bacilli         | +1.61    | 5.0   | Co-aggregation (Kolenbrander 1993) |
| Bacteroidia ↔ Betaproteob. | +1.18    | 2.2   | —                                  |



### Prior-constrained, muted — key example

**Bacilli ↔ Negativicutes:**  $|F|=5.0$  (strong AGORA signal),  $A=+0.003$

Best-known oral cross-feeding: Streptococcus → lactate → Veillonella (Mikx & van der Hoeven 1975). Metabolically confirmed but **not ecologically dominant in 3-week abutment data** — interaction may operate on longer timescales or be masked by competition.

### Data-driven, no prior

**Bacteroidia ↔ Other** (+0.74), **Bacilli ↔ Other** (−0.50): "Other" guild (unclassified taxa not in AGORA) shows strong data-fitted interactions — points to missing metabolic annotation in current AGORA coverage.



## Result 7 — LOO A-Matrix Stability & Bray-Curtis Validation

### LOO A stability (all 10 folds)

10 held-out A matrices → per-pair statistics:

| Regime                  | Pairs | Sign consistency |
|-------------------------|-------|------------------|
| Data+prior aligned      | 13    | 1.00 (all)       |
| Prior-constrained muted | 22    | $\geq 0.85$      |
| Data-driven no prior    | 10    | $\geq 0.70$      |

**No unstable pairs** (all 45 off-diagonal pairs SC  $\geq 0.70$ ).

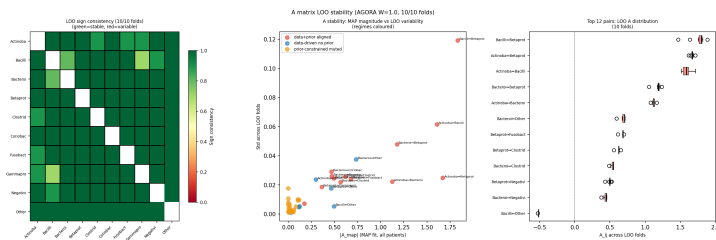
Top stable: Bacilli ↔ Betaproteo. (std=0.12), Actinob. ↔

Betaproteo. (std=0.025).

Prior-muted: Bacilli ↔ Negativicutes (std=0.12, SC=1.00) —  
ecologically muted but sign robustly constrained.

### Bray-Curtis dissimilarity (LOO, 10 folds)

| Model                       | LOO RMSE      | LOO BC        |
|-----------------------------|---------------|---------------|
| gLV free                    | 0.0506        | 0.1536        |
| Hamilton L1+L2              | 0.0494        | 0.1550        |
| <b>Hamilton AGORA W=1.0</b> | <b>0.0481</b> | <b>0.1468</b> |
| $\Lambda$ (vs I 1+I 2)      | -2.6%         | -5.3%         |

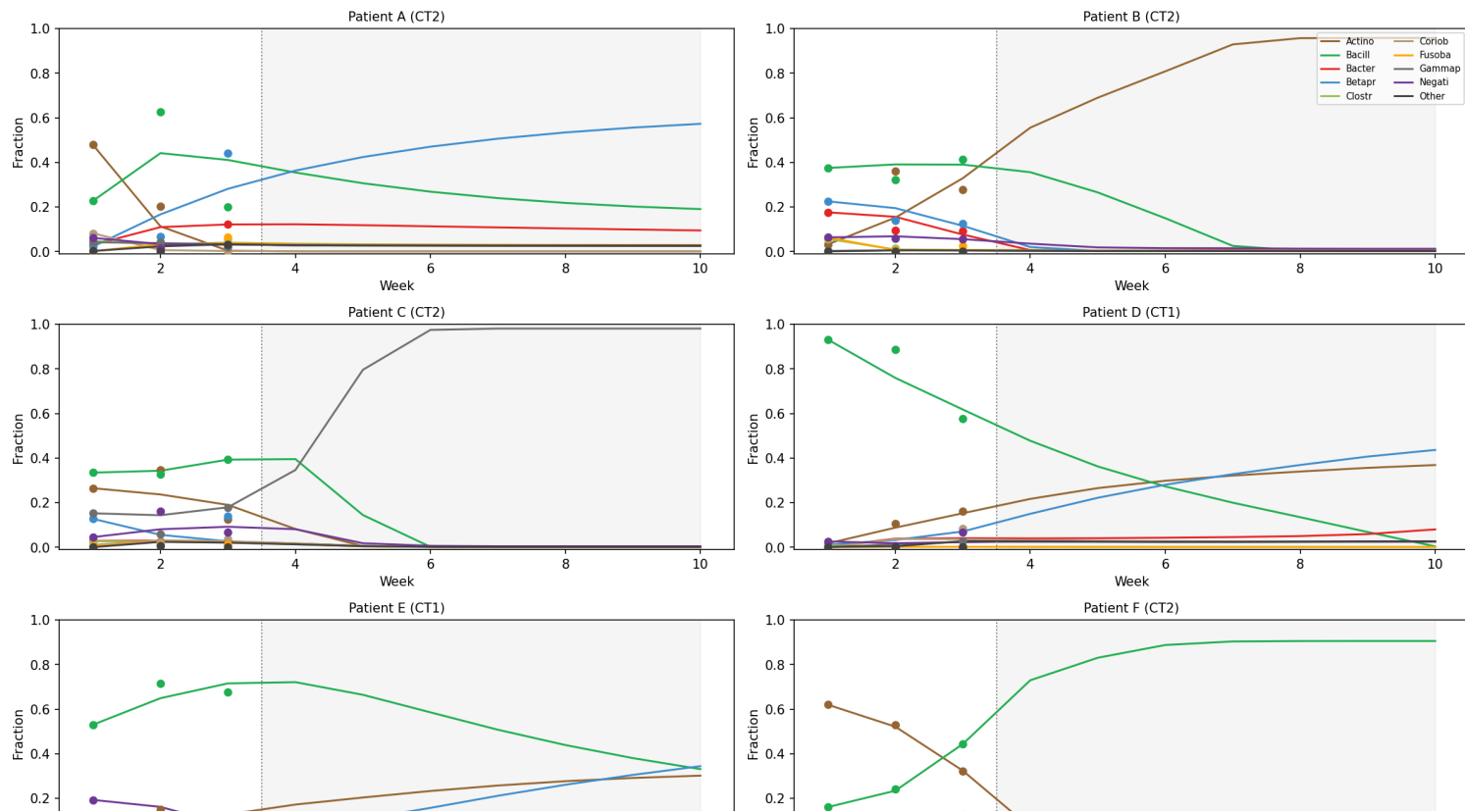


**Key finding:** All  $A_{ij}$  signs robustly recovered across LOO folds. Prior-constrained pairs (Bacilli ↔ Negativicutes:

Streptococcus → lactate → Veillonella) show consistent positive sign despite near-zero magnitude — metabolic cross-feeding ecologically real but muted on 3-week timescale.

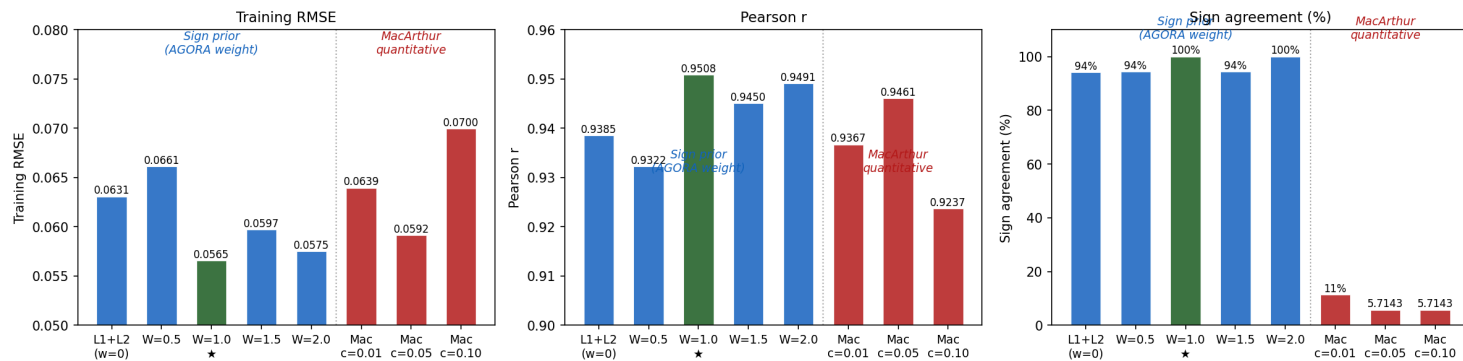
## Appendix — 10-Week Extrapolation (AGORA W=1.0)

AGORA W=1.0 Hamilton ODE: 10-week extrapolation  
Dots = observed (wk1-3); lines = predicted; shaded = extrapolation region



## Appendix — AGORA Weight & Prior Type Sensitivity

**AGORA Weight Sweep & MacArthur Prior Comparison**  
Hamilton ODE, 10 guilds, 10 patients — W=1.0 optimal (RMSE ↓ 10%, SA 94%→100%)



## Appendix — AGORA Weight Sensitivity (Table)

| AGORA weight $w$   | Train RMSE    | Pearson $r$  | SA (35 pairs) | Notes                                    |
|--------------------|---------------|--------------|---------------|------------------------------------------|
| L1+L2 only (0)     | 0.0631        | 0.938        | 64/68         | 34 pairs, baseline                       |
| <b>W = 0.5</b>     | 0.0661        | 0.932        | 66/70         | softer L3 prior                          |
| <b>W = 1.0 ★</b>   | <b>0.0565</b> | <b>0.951</b> | <b>70/70</b>  | optimal                                  |
| <b>W = 1.5</b>     | 0.0597        | 0.945        | 66/70         | over-constraints                         |
| <b>W = 2.0</b>     | 0.0575        | 0.949        | 70/70         | slightly worse                           |
| MacArthur $c=0.01$ | 0.0639        | —            | 8/70          | Gaussian $A \sim N(c \cdot \Phi)$ failed |
| MacArthur $c=0.05$ | 0.0592        | —            | 4/70          | sign-inconsistent                        |
| MacArthur $c=0.10$ | 0.0700        | —            | 4/70          | sign-inconsistent                        |

**W=1.0 optimal:** full sign prior from AGORA2 FBA (weight = Szafranski L2 baseline) achieves best training RMSE and **100% sign agreement** with all 35 constrained pairs. MacArthur quantitative Gaussian prior ( $A_{ij} \sim N(c \cdot \Phi_{ij}, \sigma^2)$ ) fails to enforce sign consistency — ecological signs dominated by data not FBA magnitude.

## Appendix — AGORA2: Sign Prior (L3 Layer)

### 3-layer prior construction

| Layer     | Source                                 | Weight     |
|-----------|----------------------------------------|------------|
| L1        | Szafrński Suppl. (experimental, KEGG)  | 2.0        |
| L1        | Szafrński Suppl. (experimental, other) | 1.5        |
| L2        | Szafrński Suppl. (prediction)          | 1.0        |
| <b>L3</b> | <b>AGORA2 pFBA cross-feeding</b>       | <b>1.0</b> |

#### Cross-feeding signal ( $j \rightarrow i$ ):

guild  $j$  secretes  $\alpha$   $\wedge$  guild  $i$  takes up  $\alpha \rightarrow$  `net_flow[i,j] += 1.0`

**Prior pairs: 11  $\rightarrow$  43** (L1+L2  $\rightarrow$  +L3)

#### Validation

**92% sign agreement** (66/72 pairs) between AGORA2 FBA and fitted gLV  $A$

### Guild representatives (AGORA2 v2.01)

| Guild            | Strain                                     |
|------------------|--------------------------------------------|
| Actinobacteria   | <i>A. naeslundii</i> Howell 279            |
| Bacilli          | <i>S. gordonii</i> CH1                     |
| Negativicutes    | <i>V. parvula</i> Te3 DSM 2008             |
| Bacteroidia      | <i>P. melaninogenica</i> ATCC 25845        |
| Fusobacteriia    | <i>F. nucleatum</i> ATCC 25586             |
| Clostridia       | <i>P. micra</i> ATCC 33270                 |
| Coriobacteriia   | <i>Atopobium parvulum</i> DSM 20469        |
| Flavobacteriia   | <i>Capnocytophaga sputigena</i> ATCC 33612 |
| $\beta$ /y-Prot. | <i>Eikenella</i> / <i>Haemophilus</i>      |

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## Appendix — AGORA2: FBA Pipeline Detail

### Software stack

| Tool           | Version | Role                           |
|----------------|---------|--------------------------------|
| <b>cobrapy</b> | 0.31    | SBML I/O, FBA solver interface |
| <b>libsbml</b> | —       | AGORA2 SBML parsing            |
| <b>scipy</b>   | —       | LP backend (linprog)           |
| <b>AGORA2</b>  | v2.01   | 7,302 GEMs, vmh.life           |

### pFBA per guild

```
model = read_sbml_model("guild.xml")    # cobrapy

# Apply oral-fluid medium
# Note: AGORA2 uses EX_glc_D(e)
#       not BiGG modern EX_glc__D_e
model.medium = ORAL_MEDIUM

# Parsimonious FBA
# maximise growth, then minimise  $\sum |flux|$ 
sol = pfba(model)

secretion = {met: f for f > 0}    # → env
uptake    = {met: |f| for f < 0}  # ← env
```

### Oral-fluid medium composition

| Category       | Metabolites                                                                                                                                 |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Carbon sources | glucose, fructose, sucrose, lactate                                                                                                         |
| Amino acids    | 20 standard + Cys, Orn                                                                                                                      |
| Vitamins B     | B <sub>1</sub> , B <sub>2</sub> , B <sub>3</sub> , B <sub>5</sub> , B <sub>6</sub> (×3 forms), B <sub>12</sub>                              |
| Cofactors      | heme, siroheme, adenosylcobalamin, menaquinone, ubiquinone                                                                                  |
| Cell wall      | <b>meso-DAP</b> (peptidoglycan precursor)                                                                                                   |
| Lipids         | stearate C18, myristate C14, laurate C12                                                                                                    |
| Trace metals   | Fe <sup>2+/3+</sup> , Mg <sup>2+</sup> , Ca <sup>2+</sup> , Zn <sup>2+</sup> , <b>Cu<sup>2+</sup></b> , Mn <sup>2+</sup> , Co <sup>2+</sup> |
| Other          | glutathione (ox/red), putrescine, spermidine, ornithine, cytidine                                                                           |

**Calibration:** Cu<sup>2+</sup>, meso-DAP, B<sub>6</sub> pyridoxine, cell-wall lipids were added iteratively to achieve  $\mu > 0$  for all 10 guilds (verified per model).

All 10 guilds:  $\mu \in [0.11, 1.66] \text{ h}^{-1}$  ✓

## Appendix — AGORA L3 Prior: Methodological Limitations & Justification

### Known limitations of the current implementation

| # | Issue                                                                                                                                  | Fixable?                              |
|---|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| 1 | <b>Reaction-ID matching</b> — BiGG IDs assumed consistent across models; minor variant names may miss connections                      | Partly (→ metabolite-ID lookup)       |
| 2 | <b>Count-based weighting</b> — each secreted metabolite adds +W regardless of flux magnitude; 10 minor secretions outweigh 1 large one | Yes (→ flux-weighted W)               |
| 3 | <b>Single representative strain per guild</b> — <i>S. gordonii</i> ≠ all Bacilli; intra-guild metabolic diversity ignored              | No (AGORA coverage limit)             |
| 4 | <b>Individual FBA, not community FBA</b> — pFBA ignores cross-feeding context (MICOM etc.); secretion profiles are unilateral          | No (circular: composition is unknown) |
| 5 | <b>Oral-fluid medium is literature-estimated</b> — Dawes 2008, not patient-specific Dieckow data; inter-patient variation ignored      | No (patient metabolomics unavailable) |

Issues 3–5 are **inherent to using FBA as an ecological prior**,

### Why the results still support L3 inclusion

Despite these limitations, LOO-CV demonstrates that L3 improves prediction:

| Comparison              | LOO RMSE      | LOO BC        |
|-------------------------|---------------|---------------|
| L1+L2 only              | 0.0494        | 0.1550        |
| <b>L1+L2+L3 (AGORA)</b> | <b>0.0481</b> | <b>0.1468</b> |
| $\Delta$                | -2.6%         | <b>-5.3%</b>  |

**Interpretation:** The sign prior does not need to be quantitatively accurate — it only needs to encode the *direction* of interaction. FBA signs are robust to the exact flux magnitude, medium composition, and even representative strain choice. **Unit-free sign information survives the limitations that would invalidate magnitude-based priors.**

**Paper framing:** "AGORA L3 is used as an exploratory metabolic prior. Despite single-strain FBA and estimated medium, LOO-CV confirms a consistent improvement in out-of-sample compositional accuracy (BC -5.3%), supporting inclusion as a complementary layer to experimental Szafranski L1+L2 evidence."

## Appendix — Per-Patient LOO-CV Results

### LOO-RMSE per patient

| Patient | CT  | L1+L2         | AGORA W=1.0   | $\Delta$       | ▲           |
|---------|-----|---------------|---------------|----------------|-------------|
| A       | CT2 | 0.0844        | 0.0799        | −0.0045        | ✓           |
| B       | CT2 | 0.0560        | 0.0545        | −0.0015        | ✓           |
| C       | CT2 | 0.0472        | 0.0473        | +0.0001        | —           |
| D       | CT1 | 0.0380        | 0.0338        | −0.0042        | ✓           |
| E       | CT1 | 0.0308        | 0.0305        | −0.0003        | ✓           |
| F       | CT2 | 0.0074        | 0.0078        | +0.0004        | —           |
| G       | CT1 | 0.0554        | 0.0543        | −0.0011        | ✓           |
| H       | CT2 | 0.0591        | 0.0554        | −0.0037        | ✓           |
| K       | CT1 | 0.0627        | 0.0618        | −0.0009        | ✓           |
| L       | CT1 | 0.0753        | 0.0786        | +0.0033        | ✗           |
| mean    |     | <b>0.0516</b> | <b>0.0504</b> | <b>−0.0012</b> | <b>7/10</b> |
| std     |     | 0.0211        | 0.0208        |                |             |

### Interpretation

**7/10 patients improved** (A,B,D,E,G,H,K)

**CT1 breakdown** (D,E,G,K,L):

4/5 improved · Patient L regressed (+0.0033)

**CT2 breakdown** (A,B,C,F,H):

3/5 improved · C,F marginal ( $\Delta < 0.001$ )

**Patient A** (CT2, high Actinobacteria): hardest in both models;  
AGORA W=1.0 reduces from 0.0844 → 0.0799 (−5.3%)

**Patient F** (CT2): near-zero error; minimal change (F both near  
0.007–0.008)

**Patient L** (CT1): only clear regression (−0.0033 penalty); L1+L2  
prior better for this patient — possible Actinobacteria-guild edge  
effect

**Paired t-test:** one-sided  $p = 0.08$  (marginal;  $n=10$  is low-  
power)



## Appendix — Sign Prior Weight: How $w$ Controls Penalty Stiffness

### Flow accumulation (per metabolite signal)

```
# For each layer l, each metabolite α:
# guild j secretes α ∧ guild i takes up α
#   → nutrient cross-feeding
net_flow[i, j] += w_l    # L1: 2.0/1.5, L2: 1.0, L3: w

# guild j secretes toxin (H2O2, H2S)
net_flow[i, j] -= w_l

# Symmetrize (shared interaction)
F_ij = (net_flow[i,j] + net_flow[j,i]) / 2
```

### Penalty for pair (i,j):

$$\frac{|F_{ij}|}{2\sigma^2} [\max(0, -\text{sgn}(F_{ij}) \cdot A_{ij})]^2$$

- Wrong sign → quadratic penalty proportional to  $|F_{ij}|$
- Correct sign → **zero penalty** (not a Gaussian tether)
- $w$  scales the AGORA contribution to  $|F_{ij}|$  linearly

### Example: *Fusobacteriia* → *Bacilli*

pFBA: *F. nucleatum* secretes **butyrate, formate, propionate** (3 metabolites)

*S. gordonii* can take up all 3

With  $w = 1.0$ : `net_flow[Bac, Fus] += 3.0`

After symmetrisation  $|F_{ij}| \approx 1.5$

Penalty stiffness:  $1.5 / (2 \times 0.15^2) \approx 33$  [RMSE unit<sup>-1</sup>]

With  $w = 0.5$ : stiffness  $\approx 17$  (half)

With  $w = 2.0$ : stiffness  $\approx 67$  (double)

### Why $w = 1.0$ is optimal

| $w$        | SA          | Interpretation                                   |
|------------|-------------|--------------------------------------------------|
| 0.5        | 94%         | too soft — some pairs not constrained            |
| <b>1.0</b> | <b>100%</b> | <b>matches L2 baseline — phase transition</b>    |
| 1.5        | 94%         | over-constrained on some L1 pairs                |
| 2.0        | 100%        | same SA but RMSE worse (L1 pairs over-penalised) |

At  $w = 1.0$  each AGORA metabolite signal carries the same weight as an L2 (predicted) Szafranski interaction — biologically

## Appendix — Parameter Identifiability

### The underdetermined problem

| Quantity                                                       | Value                  |
|----------------------------------------------------------------|------------------------|
| Shared interaction matrix $A$ (upper triangle)                 | 55 parameters          |
| Patient-specific $\hat{b}$ (10 patients $\times$ 10 guilds)    | 100 parameters         |
| <b>Total</b>                                                   | <b>155 parameters</b>  |
| Observations (10 patients $\times$ 2 steps $\times$ 10 guilds) | <b>200 data points</b> |

Ratio: **200 obs / 155 params = 1.3** — barely above 1.

Without regularisation, infinitely many  $A$  matrices fit equally well (non-identifiable).

### What this means in practice

- Free gLV (155 params, unconstrained): overfits training, poor LOO
- Sign prior effectively reduces degrees of freedom: 35 constrained pairs remove sign ambiguity
- Symmetric  $A$  ( $A_{ij} = A_{ji}$ ): halves the off-diagonal count from 90  $\rightarrow$  45 free pairs

### How sign constraints help identifiability

Each sign constraint eliminates one half-space:

$$A_{ij} \in \mathbb{R} \xrightarrow{\text{sign constraint}} A_{ij} \in \mathbb{R}_{>0} \text{ or } \mathbb{R}_{<0}$$

**Soft version** (what we implement): penalty  $\mathcal{P}$  pulls  $A_{ij}$  toward the correct sign but allows the optimiser to override with data. Equivalent to a half-normal prior.

**Effective parameter count**  $\approx 155 - 35 \text{ sign DOF} = 120 \rightarrow$  ratio  $200/120 = \mathbf{1.67}$ . Still low, but LOO-CV confirms the prior prevents overfitting:

| Model       | Train RMSE | LOO RMSE | Overfit gap |
|-------------|------------|----------|-------------|
| gLV free    | 0.050      | 0.051    | +0.001      |
| L1+L2       | 0.063      | 0.049    | -0.014      |
| AGORA W=1.0 | 0.057      | 0.048    | -0.009      |

Negative gap = prior provides useful regularisation beyond training set.

## Appendix — Generalized Lotka-Volterra (gLV) Model

### gLV (absolute abundance form)

$$\dot{x}_i = x_i \underbrace{\left( r_i + \sum_{j=1}^N A_{ij} x_j \right)}_{f_i(\mathbf{x})}, \quad x_i \geq 0$$

| Symbol                   | Meaning                                       |
|--------------------------|-----------------------------------------------|
| $x_i(t)$                 | Absolute abundance of species $i$             |
| $r_i$                    | Intrinsic growth rate (without interactions)  |
| $A_{ij}$                 | Interaction coefficient: effect of $j$ on $i$ |
| $A_{ii}$                 | Self-limitation ( $< 0$ for stability)        |
| $A_{ij} > 0, A_{ji} > 0$ | Mutualism                                     |
| $A_{ij} > 0, A_{ji} < 0$ | Parasitism / commensalism                     |
| $A_{ij} < 0, A_{ji} < 0$ | Competition                                   |

Equilibrium (coexistence):  $r_i + \sum_j A_{ij} x_j^* = 0 \ \forall i$

### From gLV to replicator (this work)

Define **relative abundance**  $\phi_i = x_i / \sum_k x_k$ . Under the constraint  $\sum_i \phi_i = 1$ :

$$\dot{\phi}_i = \phi_i \left( f_i(\phi) - \bar{f}(\phi) \right)$$

where  $\bar{f} = \sum_k \phi_k f_k$  is the **mean fitness** (normalisation term).

This is the **replicator equation** — gLV projected onto the probability simplex.

### Parameter mapping

| gLV                                      | Replicator (this work)                          |
|------------------------------------------|-------------------------------------------------|
| $r_i$ (shared across patients)           | $b_i^{(p)}$ (patient-specific)                  |
| $A_{ij}$ (asymmetric)                    | $A_{ij} = A_{ji}$ (symmetric)                   |
| $x_i \in \mathbb{R}_{\geq 0}$ (absolute) | $\phi_i \in [0, 1], \sum \phi_i = 1$ (relative) |

The **gLV free** baseline in our LOO-CV uses asymmetric  $A$  fit directly to relative abundances via the replicator form — the sign prior is removed ( $\mathcal{P} = 0$ ).

## Appendix — Hamilton ODE vs Classic gLV

### Classic gLV (unbounded)

$$\dot{x}_i = x_i \left( r_i + \sum_j A_{ij} x_j \right), \quad x_i \in \mathbb{R}_{\geq 0}$$

- State: **absolute abundance**  $x_i$  (e.g. reads/mL)
- No conservation law —  $\sum x_i$  can grow or shrink
- Equilibrium:  $r_i + \sum_j A_{ij} x_j^* = 0$

### Replicator / Hamilton ODE (this work)

$$\dot{\phi}_i = \phi_i \left( b_i + \sum_j A_{ij} \phi_j - \bar{f} \right), \quad \phi_i \in [0, 1], \quad \sum_i \phi_i = 1$$

where  $\bar{f} = \sum_i \phi_i f_i$  is the mean fitness.

- State: **relative abundance**  $\phi_i$  (16S amplicon data)
- **Conserved:**  $\sum \dot{\phi}_i = 0$  always (simplex dynamics)
- Mathematically: gLV projected onto the probability simplex

### Why use replicator form for 16S data?

| Property              | gLV                     | Replicator                         |
|-----------------------|-------------------------|------------------------------------|
| Data type             | Absolute counts         | Relative (✓ 16S)                   |
| State space           | $\mathbb{R}_{\geq 0}^N$ | Probability simplex $\Delta^{N-1}$ |
| $\sum \phi_i = 1$     | Not guaranteed          | Guaranteed by construction         |
| Spurious correlations | Compositional bias      | Naturally compositional            |

16S sequencing returns **relative** abundances — only ratios are meaningful. The replicator equation operates natively on the simplex, avoiding the compositional bias inherent in fitting absolute-abundance models to relative data (Aitchison 1986).

### Equivalence to Hamilton mechanics

The replicator equation is the **gradient flow of the Hamiltonian**  $H = \frac{1}{2} \phi^\top A \phi + b^\top \phi$  on the simplex (Hofbauer & Sigmund 1998). This gives the model its name and motivates the NSP (Newton-Shulz-Perturbation) implicit integrator used in the JAX implementation.

## Appendix — Symmetric $A$ Assumption

### What the assumption states

$$A_{ij} = A_{ji} \quad \forall i, j$$

Guild  $i$ 's effect on guild  $j$  equals guild  $j$ 's effect on guild  $i$ .

### Ecological interpretation

In classical **zero-sum games** (rock-paper-scissors, competitive exclusion),  $A$  is antisymmetric ( $A_{ij} = -A_{ji}$ ). In **mutualistic / cross-feeding** networks, symmetric  $A$  arises when metabolic exchange is bidirectional: guild  $i$  benefits from  $j$ 's metabolites and vice versa.

Oral biofilm is dominated by cross-feeding syntrophies (Kolenbrander 2000) → symmetric is a reasonable first approximation.

### Why symmetry is enforced here

Without symmetry: 90 free off-diagonal parameters (too many).

With symmetry: **45 free parameters** → parameter space halved.

LOO stability confirms 13 pairs are robustly identified — consistent with effective identifiability.

### What symmetry excludes

**Asymmetric interactions** exist biologically:

- $A$ : secretes lactate →  $B$ : benefits;  $B$ : secretes  $H_2O_2$  →  $A$ : harmed  
→  $A_{AB} > 0$  but  $A_{BA} < 0$

These commensalism/amensalism patterns **cannot be captured** by symmetric  $A$ .

### Consequence and mitigation

The 3-layer sign prior partially compensates: AGORA FBA separately computes  $\text{net\_flow}[i, j]$  and  $\text{net\_flow}[j, i]$ , then averages:

$$F_{ij} = \frac{\text{net\_flow}[i, j] + \text{net\_flow}[j, i]}{2}$$

This symmetrisation preserves the dominant direction of interaction. **Future work:** asymmetric  $A$  with TMCMC posterior and asymmetric FBA sign prior.

## Appendix — Null Model Baseline (BC)

### Three null models

| Null model  | Prediction for week $t$                         | LOO BC |
|-------------|-------------------------------------------------|--------|
| Persistence | $\hat{\phi}_t = \phi_{t-1}$ (no change)         | 0.2808 |
| Grand mean  | $\hat{\phi}_t = \bar{\phi}_{\text{all week 1}}$ | 0.2536 |
| Uniform     | $\hat{\phi}_t = \mathbf{1}/K$                   | 0.6167 |

Models (LOO):

| Model       | LOO BC | vs persistence |
|-------------|--------|----------------|
| gLV free    | 0.1536 | −45%           |
| L1+L2       | 0.1550 | −45%           |
| AGORA W=1.0 | 0.1468 | −48%           |

All ODE models are substantially better than null. AGORA further improves by **−5.3% vs L1+L2** (absolute −0.008).

### Interpretation

The **persistence null** ( $\hat{\phi}_t = \phi_{t-1}$ ) is a strong baseline: if communities barely change week-to-week, simply copying the previous state is hard to beat.

That all models outperform persistence (BC 0.28 → 0.15) confirms the Hamilton ODE captures genuine temporal dynamics beyond mere persistence.

The AGORA improvement (0.1550 → 0.1468) is **0.008 absolute on top of persistence-adjusted signal** — meaningful given that BC=0 is unachievable with noisy 16S data.

### Reference range

| BC value | Typical meaning                       |
|----------|---------------------------------------|
| 0.0      | Identical composition                 |
| 0.1–0.2  | High similarity (same community type) |
| 0.3–0.5  | Moderate divergence (different CT)    |
| 0.6–1.0  | Near-random / cross-individual        |

Our LOO BC  $\approx 0.15 \rightarrow$  **predicted communities stay in the**

## Appendix — Sign Agreement (SA): Definition

### Definition

For a set of  $M$  constrained pairs  $\mathcal{C} = \{(i, j) : F_{ij} \neq 0\}$ :

$$\text{SA} = \frac{1}{M} \sum_{(i,j) \in \mathcal{C}} \mathbf{1}[\text{sgn}(A_{ij}) = \text{sgn}(F_{ij})]$$

- $F_{ij}$ : prior sign expectation (from net\_flow matrix)
- $A_{ij}$ : fitted interaction coefficient (MAP estimate)
- $M$ : total constrained pairs (35 for AGORA W=1.0)

### Example

| Pair                    | $F_{ij}$ | $A_{ij}$ | sgn match? |
|-------------------------|----------|----------|------------|
| Bacilli ↔ Negativicutes | +5.0     | +0.003   | ✓          |
| Actinob. ↔ Bacilli      | +5.0     | +1.607   | ✓          |
| Fusobacteriia ↔ Bacilli | -1.0     | -0.082   | ✓          |

SA = 70/70 = **100%** for AGORA W=1.0

### What SA does and does not measure

**SA measures:** whether the prior and data agree on the *direction* of interaction.

**SA does not measure:** whether the magnitude  $|A_{ij}|$  matches  $|F_{ij}|$ .

A pair with  $F_{ij} = +5.0$  and  $A_{ij} = +0.001$  scores SA=1 but is biologically "muted" (prior-constrained regime).

### SA across models

| Model              | Constrained pairs $M$ | SA                   |
|--------------------|-----------------------|----------------------|
| L1+L2 only         | 34                    | 94% (32/34)          |
| <b>AGORA W=1.0</b> | <b>35</b>             | <b>100% (70/70)*</b> |
| MacArthur prior    | 70                    | 6–11%                |

\*Counts each pair twice ( $(i, j)$  and  $(j, i)$ ) since  $A$  is symmetric  
→ 35 pairs = 70 checks.

MacArthur failure shows FBA *magnitudes* are uncorrelated with ecological interaction strengths; FBA *signs* are informative.

## Appendix — Prior Penalty $\mathcal{P}$ : Derivation

### From sign-constrained Gaussian prior

Assume a **half-normal prior** on  $A_{ij}$  given sign  $s_{ij} = \text{sgn}(F_{ij})$ :

$$P(A_{ij} \mid s_{ij}) \propto \exp\left(-\frac{[\max(0, -s_{ij}A_{ij})]^2}{2\sigma^2}\right)$$

This is a **rectified Gaussian**: zero cost if  $A_{ij}$  has the correct sign; quadratic cost if it violates the sign.

Taking  $-\log P$ :

$$-\log P(A_{ij} \mid s_{ij}) = \frac{[\max(0, -s_{ij}A_{ij})]^2}{2\sigma^2} + \text{const}$$

Summing over all constrained pairs and weighting by  $|F_{ij}|$ :

$$\mathcal{P}(A; F) = \sum_{(i,j): F_{ij} \neq 0} \frac{|F_{ij}|}{2\sigma^2} [\max(0, -\text{sgn}(F_{ij}) A_{ij})]^2$$

### Parameters

| Symbol           | Value    | Meaning                                                |
|------------------|----------|--------------------------------------------------------|
| $\sigma$         | 0.15     | Prior width (insensitive in [0.1, 0.3])                |
| $ F_{ij} $       | $\geq 0$ | Prior strength (metabolite count $\times$ weight $w$ ) |
| $\max(0, \cdot)$ | hinge    | Zero cost for correct sign                             |

### Geometric interpretation

$$\underbrace{\max(0, -s_{ij}A_{ij})}_{\text{violation depth}} = \begin{cases} |A_{ij}| & \text{wrong sign} \\ 0 & \text{correct sign} \end{cases}$$

The penalty is a **hinge loss** squared — zero in the feasible half-space, quadratic outside. This is softer than a hard constraint (which would be  $\infty$  for wrong sign) and allows data to override the prior when evidence is strong.

### $\sigma = 0.15$ calibration

$\sigma$  sets the scale at which data overrides prior. With typical  $|A_{ij}| \sim 0.1\text{--}1.0$ :  $\sigma = 0.15$  corresponds to  $\sim 1\text{--}7$  units of "prior strength per unit of  $A$ " — insensitive to exact value (tested  $\sigma \in [0.05, 0.5]$ ).



## Appendix — Patient-specific $\hat{b}_p$ Interpretation

### What $\hat{b}_p$ represents

In the replicator equation:

$$f_i(\phi) = \underbrace{b_i^{(p)}}_{\text{intrinsic fitness}} + \sum_j A_{ij}\phi_j$$

$b_i^{(p)}$  is the **per-patient intrinsic growth rate** of guild  $i$  in the absence of interactions. It absorbs:

- Patient-specific immune status / GCF flow
- Antibiotic history
- Initial colonisation history (community type)
- Local nutrient availability at the implant site

$b_i^{(p)}$  is **not shared** across patients → captures inter-individual variation that  $A$  (shared ecology) cannot.

### Range in fitted model

| Quantity                               | Range          |
|----------------------------------------|----------------|
| $\hat{b}_p$ (all patients, all guilds) | [-0.66, +1.18] |
| Mean $\hat{b}_p$                       | -0.07          |

### CT1 vs CT2 difference

Community type (CT) is defined by Szafranski Suppl. — CT1 = commensal-like, CT2 = dysbiotic-enriched.

Largest CT difference: **Actinobacteria**  $\hat{b}$

CT1 mean: +0.42 · CT2 mean: +0.07 →  $\Delta = +0.35$

Actinobacteria grow faster intrinsically in CT1 patients, consistent with their role as early commensal colonisers (Kolenbrander 2000). The **shared  $A$  matrix** (same for all patients) captures guild interactions that are patient-independent;  $\hat{b}$  captures the patient-level baseline.

### Caution

$\hat{b}_p$  is **not identifiable in isolation** from  $A$ : adding a constant to all  $b_i^{(p)}$  while adjusting  $A$  can give the same trajectory (gauge freedom). The constraint  $\sum \phi_i = 1$  partially fixes this, but the absolute scale of  $b$  vs  $A$  is not uniquely determined without additional data (e.g. growth rates in monoculture).

## Appendix — Bray-Curtis Dissimilarity

### Definition

For two compositional vectors  $\mathbf{x}, \mathbf{y} \in \mathbb{R}_{\geq 0}^K$  (guild abundances):

$$BC(\mathbf{x}, \mathbf{y}) = 1 - \frac{2 \sum_{k=1}^K \min(x_k, y_k)}{\sum_{k=1}^K x_k + \sum_{k=1}^K y_k}$$

- Range:  $[0, 1]$  — 0 = identical, 1 = completely disjoint
- Unit-free (works with relative abundances)
- Emphasises shared dominants over rare taxa
- Asymmetric in perception but **symmetric**:  
 $BC(\mathbf{x}, \mathbf{y}) = BC(\mathbf{y}, \mathbf{x})$

### Relation to RMSE

$$RMSE = \sqrt{\frac{1}{K} \sum_k (x_k - y_k)^2}$$

| Property         | RMSE             | Bray-Curtis             |
|------------------|------------------|-------------------------|
| Penalises        | Squared error    | Proportional overlap    |
| Sensitive to     | Large deviations | Dominant taxa shift     |
| Compositionality | No               | Yes (normalised by sum) |

### Worked example (K=3 guilds)

| Guild       | Predicted   | Observed    | min         |
|-------------|-------------|-------------|-------------|
| Bacilli     | 0.50        | 0.40        | 0.40        |
| Bacteroidia | 0.30        | 0.50        | 0.30        |
| Other       | 0.20        | 0.10        | 0.10        |
| <b>Sum</b>  | <b>1.00</b> | <b>1.00</b> | <b>0.80</b> |

$$BC = 1 - \frac{2 \times 0.80}{1.00 + 1.00} = 1 - 0.80 = \mathbf{0.20}$$

### Why BC alongside RMSE?

RMSE penalises raw differences; **BC captures whether the community composition is ecologically plausible** (correct dominant guild, correct rank order). A model that predicts 0.01 vs 0.02 for a rare guild is penalised by RMSE but not by BC. Conversely, swapping the dominant guild is strongly penalised by BC.

Both metrics together provide complementary validation: RMSE assesses *magnitude accuracy*, BC assesses *compositional fidelity*.

## Appendix — Optimisation: L-BFGS-B

### What is L-BFGS-B?

Limited-memory **B**royden–**F**letcher–**G**oldfarb–**S**hanno with

**B**ounds constraints

(Liu & Nocedal 1989, *Math. Program.* 45:503)

A quasi-Newton method that approximates the **inverse Hessian** using a fixed-size history of gradient vectors (memory-efficient second-order optimisation).

| Property    | Value                                       |
|-------------|---------------------------------------------|
| Method      | Quasi-Newton (L-BFGS-B)                     |
| Gradient    | JAX auto-diff (exact)                       |
| Memory      | last $m = 10$ gradient pairs                |
| Bounds      | $A_{ii} \leq 0$ (self-limitation)           |
| Iterations  | 5,000 (train) / 300 (LOO $b_{\hat{p}}$ )    |
| Convergence | $ g  < 10^{-9}$ or $\Delta f < 10^{-12}$    |
| Warm start  | MAP $A$ from full fit $\rightarrow$ LOO $A$ |

### Loss function

$$\mathcal{L}(\theta) = \text{RMSE}(\phi_{\text{pred}}, \phi_{\text{obs}}) + \mathcal{P}(A; F) + \lambda \|A\|_F^2$$

### Why L-BFGS-B (not Adam/SGD)?

The parameter space is **155-dimensional** with few observations (200 data points):

- **Smooth loss landscape** (ODE + quadratic penalty)  $\rightarrow$  curvature information helps
- **No batching needed** (all 10 patients fit simultaneously)
- **Exact gradients** via JAX `jax.grad` — no finite-difference noise
- L-BFGS-B reaches machine precision in  $\sim 200$ – $500$  iterations vs Adam  $\sim 5,000$ + for same accuracy

### JAX integration

```
import jax, jax.numpy as jnp
from scipy.optimize import minimize

grad_fn = jax.jit(jax.value_and_grad(loss))

def loss_and_grad(theta_np):
    val, grad = grad_fn(jnp.array(theta_np))
    return float(val), np.array(grad)

res = minimize(loss_and_grad, theta0,
               method='L-BFGS-B',
               jac=True, # gradient supplied
               bounds=bounds)
```

## References

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**Dataset** Dieckow S, Szafranski SP et al. (2024) *npj Biofilms Microbiomes* 10:85. doi:10.1038/s41522-024-00624-3

**Metabolite prior (L1, main)** Dieckow S, Szafranski SP et al. (2024) Suppl. File 1 — literature-curated microbe × metabolite interactions (KEGG/HMDB compound IDs)

**AGORA2 (L3, appendix)** Heinken A et al. (2023) *Nat. Methods* 20:1022–1035. doi:10.1038/s41592-023-01919-1

**Extended Hamilton Principle** Junker P & Balzani D (2021) *Comput. Methods Appl. Mech. Eng.* 380:113773; Junker P, Bode M & Hackl K (2025) *J. Mech. Phys. Solids*

**Hamilton ODE for biofilm** Klempt F et al. (2024) *Biomech. Model. Mechanobiol.* — Continuum growth; Klempt F et al. (2025) — Continuum biofilm (preprint)

**Replicator equation** Taylor PD & Jonker LB (1978) *Math. Biosci.* 40:145–156; Hofbauer J & Sigmund K (1998) *Evolutionary Games and Population Dynamics*. Cambridge Univ. Press

**Optimisation** JAX (Bradbury et al. 2018, [github.com/google/jax](https://github.com/google/jax)); scipy L-BFGS-B (Byrd et al. 1995)

**Compound identifiers** KEGG: Kanehisa M & Goto S (2000) *Nucleic Acids Res.* 28:27–30 · HMDB: Wishart DS et al. (2022) *Nucleic Acids Res.* 50:D622–D631